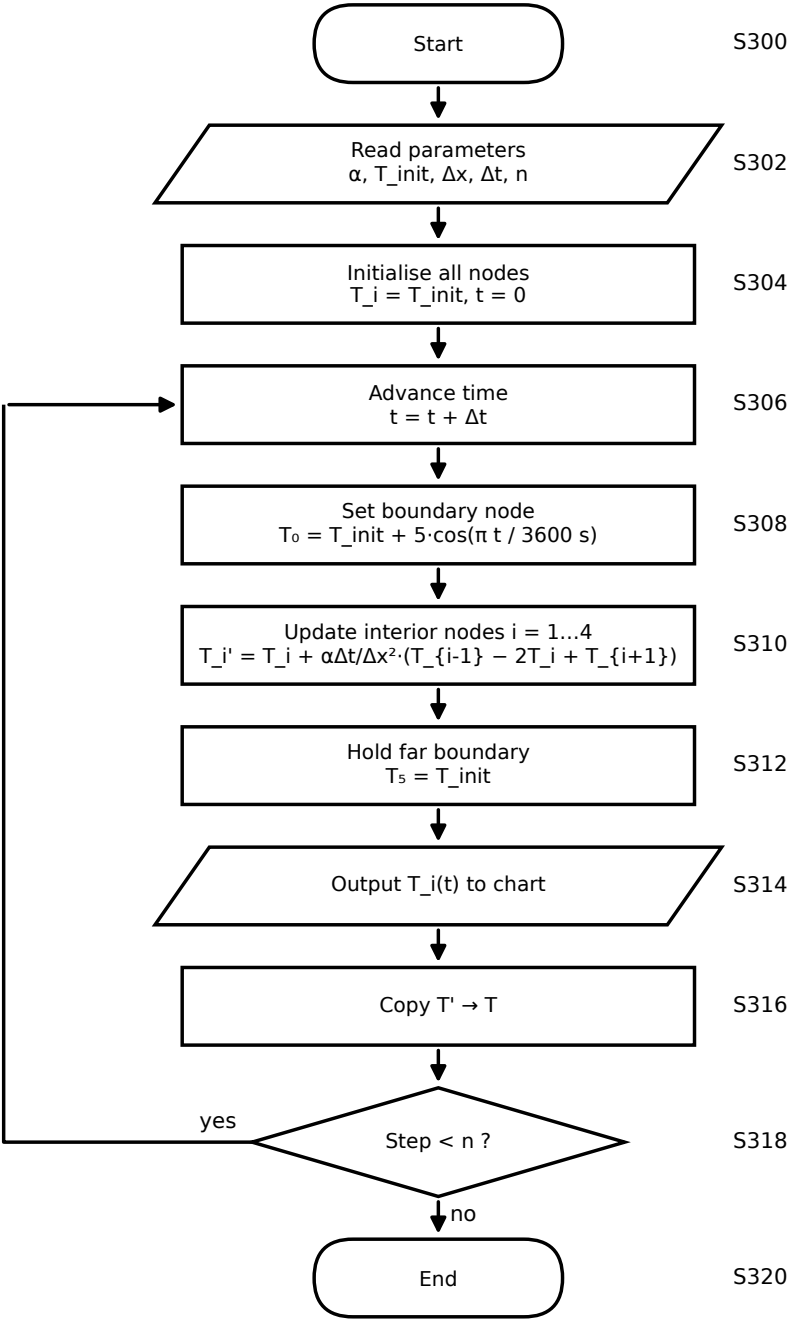
Start
S300
Read parameters
α, T_init, Δx, Δt, n
S302
Initialise all nodes
T_i = T_init, t = 0
S304
Advance time
t = t + Δt
S306
Set boundary node
T₀ = T_init + 5·cos(π t / 3600 s)
S308
Update interior nodes i = 1...4
T_i' = T_i + αΔt/Δx²·(T_{i-1} − 2T_i + T_{i+1})
S310
Hold far boundary
T₅ = T_init
S312
Output T_i(t) to chart
S314
Copy T' → T
S316
yes
Step < n ?
S318
no
End
S320

FIG. 4